

Topic 5: Few-Shot Prompting

5.1 What is Few-Shot Prompting?

- Few-shot: providing 2-8 examples of input-output pairs before the actual task
- Examples teach the model the exact pattern, format, and style you expect
- Dramatically improves consistency and accuracy for non-standard tasks
- One-shot: a single example; few-shot: 2-8 examples; many-shot: 8+ examples
- The model learns the task from examples without any weight updates -- this is called in-context learning

5.2 Structure of Few-Shot Prompts

- Pattern: [instruction] + [example 1 input -> output] + [example 2 input -> output] + [actual input]
- Examples should cover the range of cases you expect in real inputs
- Keep examples consistent in format -- any deviation will confuse the model
- Always end with the actual input and a prompt for the model to fill in the output

```
Classify the tone of each sentence as: Formal, Casual, or Aggressive.
```

```
Input: 'Please review the attached document at your earliest convenience.'
```

```
Output: Formal
```

```
Input: 'Hey, check this out when you get a sec!'
```

```
Output: Casual
```

```
Input: 'I need this done NOW, no excuses.'
```

```
Output: Aggressive
```

```
Input: 'Kindly ensure compliance with the stated regulations.'
```

```
Output:
```

5.3 Choosing Good Examples

- Representative: examples should reflect the actual distribution of real inputs
- Diverse: cover edge cases, not just the easy ones
- Correct: one wrong example can mislead the whole task -- double-check all outputs
- Consistent: same format, same label style, same level of detail across all examples
- Order matters: the last 1-2 examples have the most influence -- put strong examples last

5.4 Few-Shot for Custom Formats

- Few-shot is ideal when you need a custom output structure the model wouldn't produce by default
- Use it for: custom JSON schemas, domain-specific labels, company-specific writing styles
- Combine with explicit format instructions for best results

Extract product info in this exact format:

Input: 'The Nike Air Max 90 in size 10 costs \$120 and is available in red and black.'

Output: {name: 'Nike Air Max 90', size: 10, price: 120, colors: ['red', 'black']}

Input: 'Adidas Ultraboost 22, size 9, \$180, comes in white and grey only.'

Output: {name: 'Adidas Ultraboost 22', size: 9, price: 180, colors: ['white', 'grey']}

Input: 'Puma RS-X size 11, \$95, sold in blue, yellow, and green.'

Output: